Salim Ansari 24070521005 A1

*pract10.l

```
Name - Salim Ansari
PRN:- 24070521005
Batch A1

%{
#include "y.tab.h"
%}

%%
"for"   { return FOR; }
[a-zA-Z]+  { return ID; }
[0-9]+   { return NUM; }
"<="|">="|"=="|"!="|"<"|">" { return RELOP; }
"++"  { return INC; }
"--"   { return DEC; }
[ \t\n]+  ;
.    { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```

*pract10.y

```
Name: Salim Ansari
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Batch A1

%{
#include <stdio.h>

int yylex(void);
int yyerror(const char *s);
%}

%token FOR ID NUM RELOP INC DEC

%%
input : stmt
      ;

stmt  : FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID INC ')'
      | FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID DEC ')'
      | ID RELOP ID '?' ID ':' ID
      | ID RELOP NUM '?' ID ':' ID
      | ID RELOP ID '?' NUM ':' NUM
      | ID RELOP NUM '?' NUM ':' NUM
      ;
%%

int yyerror(const char *s)
{
    printf("Invalid statement\n");
    return 1;
}

int main()
{
    printf("Enter a for loop or ternary expression:\n");

    if(yyparse() == 0)
        printf("Valid statement\n");

    return 0;
}
```

```
salim@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0;i<25;i++)
Valid statement
salim@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>g?a:g
Valid statement
salim@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0,i<n,i++)
Invalid statement
salim@LAPTOP-ICU7078D:~$ ./a.out
Enter a for loop or ternary expression:
a>b!a:b
Invalid statement
salim@LAPTOP-ICU7078D:~$
```

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